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Class : TE_A

Batch : A3

Roll No : 13165

[5]:

```
import pandas as pd
df=pd.read_csv(r"C:\Users\CNLAB07\Desktop\DSBDA\iris.csv")
```

[7]:

```
df
```

[7]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	\
0	1	5.1	3.5	1.4	0.2	
1	2	4.9	3.0	1.5	0.2	
2	3	4.7	3.2	1.3	0.2	
3	4	4.6	3.1	1.5	0.2	
4	5	5.0	3.6	1.4	0.2	
	...	...	...	...	...	
145	146	6.7	3.0	5.2	2.3	
146	147	6.3	2.5	5.0	1.9	
147	148	6.5	3.0	5.2	2.0	
148	149	6.2	3.4	5.4	2.3	
149	150	5.9	3.0	5.1	1.8	

	Species
0	Iris-setosa
1	Iris-setosa
2	Iris-setosa
3	Iris-setosa
4	Iris-setosa
	...
145	Iris-virginica
146	Iris-virginica
147	Iris-virginica
148	Iris-virginica
149	Iris-virginica

[150 rows x 6 columns]

[13]:

```
df.groupby(['Species'])['PetalLengthCm'].mean()
```

[13]: Species

Iris-setosa	1.464
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```
Iris-versicolor    4.260
Iris-virginica     5.552
Name: PetalLengthCm, dtype: float64
```

```
[15]: df.groupby(['Species'])['PetalLengthCm'].std()
```

```
[15]: Species
Iris-setosa      0.173511
Iris-versicolor  0.469911
Iris-virginica   0.551895
Name: PetalLengthCm, dtype: float64
```

```
[17]: df.groupby(['Species'])['SepalLengthCm'].std()
```

```
[17]: Species
Iris-setosa      0.352490
Iris-versicolor  0.516171
Iris-virginica   0.635880
Name: SepalLengthCm, dtype: float64
```

```
[19]: df.groupby(['Species'])['SepalLengthCm'].mean()
```

```
[19]: Species
Iris-setosa      5.006
Iris-versicolor  5.936
Iris-virginica   6.588
Name: SepalLengthCm, dtype: float64
```

```
[21]: import numpy as np
```

```
[25]: sort1=sorted(df['SepalLengthCm'])
      np.percentile(sort1,25)
```

```
[25]: 5.1
```

```
[27]: sort1=sorted(df['SepalWidthCm'])
      np.percentile(sort1,25)
```

```
[27]: 2.8
```

```
[29]: sort1=sorted(df['PetalLengthCm'])
      np.percentile(sort1,25)
```

```
[29]: 1.6
```

```
[31]: sort1=sorted(df['PetalWidthCm'])
      np.percentile(sort1,25)
```

[31]: 0.3

[]: